

Partho Kumar Paul, PhD candidate, Department of Chemistry

University of California, Irvine

pkpaul@uci.edu | 951-281-9240 | LinkedIn Profile: <https://www.linkedin.com/in/partho-p-a98728a1/>

Professional Summary

PhD candidate in **synthetic organic chemistry** with a strong background in designing and developing **redox-active organic molecules for CO₂ capture and release**. Experienced in both **organic and inorganic chemistry**, with expertise in **main group catalysis, transition metal ligand design, and small-molecule synthesis**. Skilled in applying **physical organic chemistry principles** to optimize molecular performance and stability.

Proficient in a variety of **spectroscopic and analytical techniques**, including **NMR, GC-MS, LC-MS, cyclic voltammetry (CV), and UV-VIS spectroscopy**, to characterize reaction mechanisms, binding affinities, and molecular structures. Passionate about **mentoring and teaching**, with several years of experience guiding undergraduate students in laboratory settings, developing instructional materials, and fostering an inclusive and engaging learning environment.

Education

PhD candidate, Organic Chemistry

UCI, expected graduation: Spring 2026

Bachelor of Science, Chemical Engineering with Nanotechnology emphasis

Minor: Chemistry

UCR, Fall 2015-Spring 2020

Work Experience

Graduate Student Researcher

University of California, Irvine, Yang Lab | April 2024 – Present

- Designed, synthesized, and characterized **15+ redox-active organic molecules** for electrochemical CO₂ capture and release in both aqueous and organic media.
- Applied **physical organic chemistry principles** to enhance CO₂ selectivity while minimizing O₂-induced deactivation.
- Utilized analytical techniques such as **NMR, UV-VIS spectroscopy, and cyclic voltammetry** to quantify CO₂ binding efficiency.
- **Mentored undergraduate students**, providing guidance on research techniques and laboratory best practices.

Teaching Assistant, Chemistry

University of California, Irvine | August 2021 – Present

- Led discussion sections and lab courses for undergraduate chemistry students, simplifying complex concepts through interactive teaching methods.
- Provided individualized support through office hours, fostering student engagement and comprehension.
- Developed and graded assessments, ensuring fair evaluation and constructive feedback.
- Utilized a **multi-modal teaching approach**, including visual aids and problem-based learning, to accommodate diverse learning styles.
- Gained expertise in **science communication, curriculum development, and student mentorship**.

BLOOM Faculty Intern

Irvine Valley College, Irvine | January 2025 – Present

- Collaborate with faculty mentor to design and implement inclusive teaching strategies aimed at fostering student success.
- Developed and refined course materials to enhance student engagement and comprehension.
- Assist in curriculum planning, incorporating evidence-based pedagogical practices.
- Support students through mentorship, academic guidance, and individualized learning plans.
- Gain hands-on experience in community college instruction, emphasizing diversity, equity, and inclusion in education.

Elected Representative, School of Physical Sciences

Associated Graduate Students (AGS), University of California, Irvine | May 2024 – Present

- Represent the interests and concerns of graduate students within the School of Physical Sciences at university-wide meetings.
 - Advocated for policies and initiatives aimed at improving academic resources, funding opportunities, and student well-being.
 - Collaborated with faculty, administration, and student organizations to enhance the graduate student experience.
 - Organized and participated in events promoting professional development, networking, and community engagement.
 - Contributed to decision-making processes on issues affecting graduate education, research, and campus policies.
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Skills

- **Chemical Techniques:** Organic synthesis, air-sensitive techniques (Schlenk line, glovebox), electrochemistry (cyclic voltammetry, bulk electrolysis), spectroscopy (UV-Vis, NMR, IR), and CO₂ capture methodologies
 - **Analytical Tools:** NMR spectroscopy (¹H, ¹³C), UV-Vis, FTIR, electrochemical analysis, GC-MS
 - **Computational Chemistry:** Gaussian, Avogadro, ChemDraw, preliminary experience in DFT calculations.
 - **Software:** Origin, MestReNova, Microsoft Office Suite, Adobe Illustrator, PyMOL
 - **Programming Languages:** Python (basic), MATLAB (basic)
 - **Leadership:** Experience as a TA, mentor to undergraduate researchers, and elected representative in graduate student government (AGS)
 - **Communication:** Strong written and verbal communication skills developed through teaching, research presentations, and scientific writing
 - **Teamwork:** Collaborative experience in research groups, interdisciplinary projects, and faculty development programs
 - **Mentorship:** Guided undergraduates in laboratory research and supported student learning through tutoring and office hours
 - **Adaptability:** Effectively balanced multiple roles in research, teaching, and service; quickly adjusted to new challenges and learning environments
 - **Organization:** Managed complex research timelines, course responsibilities, and student mentorship simultaneously
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Presentation

Title: Applications of physical-organic principles in enhancing electrochemical CO₂ binding ability of Quinones.

UCI graduate student and post-doctoral researcher Seminar, Nov 2024

SoCal Electrochemistry Conference for Students, April 5, 2025

Certifications & Professional Development

- Workshop on Pulse Radiolysis, Brookhaven National Laboratory, Brookhaven, NY
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Awards & Honors

Eugene Cota-Robles Fellowship (1st and 4th year of graduate studies)– *University of California, Irvine*

Awarded prestigious fellowship supporting underrepresented students in doctoral programs.

Poster Award – UC Irvine Undergraduate Research Symposium in Chemistry (Summer 2019)

Recognized for delivering the *best research presentation* at the symposium, showcasing innovative findings in chemistry.

Bourns Foundation Engineering Scholarship (Spring 2019)

Competitive scholarship awarded to outstanding students in the *Bourns College of Engineering*, providing financial support for academic expenses.

UC Leadership Excellence through Advanced Degrees (UC LEADS) (Spring 2018)

Selected for the **UC LEADS program**, a two-year research and leadership development initiative supporting promising STEM students.

Professional Memberships

- American Chemical Society (ACS), Member, August 2020-Present

Service Experience**Graduate Student Representative**

Committee on Student Teaching and Learning Experience (CSTLE), UC Irvine, 2024–2025

Advocated for the needs of teaching assistants and graduate instructors by providing feedback on teaching practices, training resources, and policies impacting instructional roles. Worked collaboratively with faculty and administrative staff to enhance the overall teaching and learning environment across campus.

Volunteer**Sanatan Bengali Society of California**

Contributed to community events and cultural programming, supporting organizational logistics, event coordination, and outreach efforts to promote cultural heritage and engagement within the Bengali community.

References

Jenny Y. Yang, PhD

Academic Advisor

Professor, UCI Chemistry j.yang@uci.edu

Steven Nguyen, PhD

Mentor

Professor, IVC chemistry snguyen@ivc.edu